LEXICON

Exercises: Agile methods and Scrum

Part 1 is done on your own: write your answers in a document. Part 2 is a group project with 4-5 people. No code, only thinking, writing and talking.

Part 1: Check your understanding

Exercise 1

Explain in your own words why the Waterfall method often goes wrong when building software. Give two reasons.

Exercise 2

Pick two of the four values in the Agile Manifesto. For each one, give an example of what it could look like in a real team.

Exercise 3

Who does what? Write Product Owner, Scrum Master or Developers for each line.

a) Decides the order of the Product Backlog
b) Decides how to build a feature
c) Helps the team when a test server has been down for two days
d) Talks to customers about what they need
e) Coaches the team when the Daily Scrum takes 45 minutes

Exercise 4

Your team works in two-week sprints. In the middle of a sprint, the CEO comes in and wants a new feature right away. What should happen, according to Scrum?

Exercise 5

What is the difference between the Sprint Review and the Sprint Retrospective? Who takes part in each?

Exercise 6

Rewrite these backlog items as user stories (As a ..., I want ..., so that ...):

a) Add login
b) Export orders to Excel
c) Dark mode

Exercise 7

Write three acceptance criteria for this story: As a customer, I want to see my order history, so that I can buy the same thing again.

Exercise 8

Write a Definition of Done for a team of four developers. Include at least five points.

Exercise 9

A team's Daily Scrum takes 45 minutes. Everyone stands facing the manager and reports what they did. What is wrong, and what would you suggest?

Exercise 10

Scrum or Kanban? Choose for each team and explain why.

a) A team building a new mobile app
b) An IT support desk
c) A team fixing bugs in a webshop that is already live

Exercise 11

In planning poker, four developers show 2, 3, 3 and 13 for the same story. What should the team do next?

Part 2: A mini Scrum project

FitLife is a small gym. It wants an app where members can see the schedule, book classes and cancel bookings. Your group is the Scrum Team that will build it. You don't write any code: you plan the work, the way a real team would before the first sprint.

What the customer wants

"I run FitLife, a small gym with about 400 members and 5 instructors. Today members book classes by phone or on a paper list at the reception. It's chaos: classes get overbooked, people forget to cancel, and the staff spends hours on the phone. I want an app where members can see the schedule, book a class and cancel. Then my staff can spend less time on bookings."

- Users: members, instructors and reception staff.
- Most important: seeing the schedule and booking a class. Without that, nothing else matters.
- Classes: yoga, spinning, boxing and strength training. Each class has a maximum number of places, usually 12 to 20.
- When a class is full, members want to join a waiting list and get a spot automatically if someone cancels.
- Members can cancel up to 2 hours before the class. After that it counts as a no-show.
- Instructors want to see who has booked their class, and mark who actually came.
- Reception staff need to add new classes and change times.
- A reminder the day before would be nice, but it's not the most important thing.
- Mobile first: most members book on their phone.
- Payment: not now. Members pay a monthly fee at the reception.
- Deadline: a first version within two months.
- Something not covered here? Make a sensible decision as a team and write it down.

#	What	What to do
1	Roles	Choose one Product Owner and one Scrum Master. The rest are Developers. Read together what the customer wants (above).
2	Product Goal	Write the Product Goal in one sentence.
3	Product Backlog	Write 10-12 user stories. The Product Owner puts them in order, most valuable first.
4	Refinement	Add 2-3 acceptance criteria to the top five stories. Write your team's Definition of Done.
5	Planning poker	Make cards from paper (1, 2, 3, 5, 8, 13) and estimate the top stories. Split any story bigger than 13.
6	Sprint Planning	Your velocity is 15 points. Choose the stories for the first sprint and write a Sprint Goal.

#	What	What to do
7	Board	Draw the Sprint Backlog as a board with To do, In progress, Review and Done. Use paper, Trello or GitHub Projects. Set a WIP limit.
8	Change!	The customer will announce a change while you work. Discuss: what goes into the backlog, and does anything change in the sprint?
9	Sprint Review	Show your board, your Sprint Goal and your top stories to the class in 3 minutes.
10	Retrospective	As a group: what went well, what was hard, and one thing you would do differently.

Hand in: a photo or screenshot of your board, your Product Goal, your Sprint Goal, your user stories with estimates, and your Definition of Done.